

EUGEN FENG

Active DoD Secret Clearance

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TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (ES6+), C/C++, ARM C, Swift

Frameworks and Libraries: React.js, Node.js, Cube.js, HTML5, CSS3, Electron, CoreMotion, OpenCV, scikit-learn

Others: DOORS, Jira, BitBucket, Git, Gitlab, Docker, Power BI, Microsoft Access, Microsoft SQL Server, StackAI

PROFESSIONAL EXPERIENCE

BAE Systems | *Software Engineer*

August 2022 – Present

- Designed and deployed agentic AI workflows using LLMs (Meta LLaMA, Mistral) within low-code orchestration platforms (StackAI), enabling intelligent chatbot systems for onboarding and knowledge retrieval
- Built and maintained modular AI pipelines by chaining components such as prompt templates, retrieval layers, and decision logic to deliver context-aware responses
- Developed LLM-driven summarization and anomaly detection workflows, improving KPI monitoring response time by 33%
- Supplemented low-code systems with Python scripts and SQL queries for data extraction, transformation, and integration into AI pipelines
- Built a full-stack application (React + Node.js) to support structured data input, reducing manual errors by 90%
- Re-architected an internal report-generation tool from a monolithic JS implementation into a FastAPI backend with Jinja templating, improving security posture and maintainability for a concurrent internal user base of 5.
- Containerized applications using Docker and implemented GitLab CI/CD pipelines to automate build, test, and deployment processes, ensuring scalable and reliable distribution to end users
- Collaborated with stakeholders to deliver production-ready solutions, improving accessibility and usability of data analytics tools across teams
- Integrated Power BI analytics dashboards into operational workflows for major defense programs, enabling real-time decision support

General Dynamics Electric Boat | *Communication Systems Engineer I, II*

August 2020 – August 2022

- Supported development of mission-critical control systems for VIRGINIA-class submarines
- Performed system validation, testing, and risk analysis across hardware/software integrations
- Conducted environmental and system-level testing to ensure operational reliability
- Managed hardware lifecycle and obsolescence planning

University at Buffalo, Department of Electrical Engineering | *Teaching Assistant*

August – December 2019

- Assisted the professor in evaluating the proficiency of 80+ students with weekly benchmarks and individual projects
- Hosted office hours to discuss technical concepts with students to enhance their understanding of embedded systems and provided guidance on their projects in ARM C and C++

Equinix | *Sales Engineer Intern*

May – August 2019

- Managed and organized data effectively utilizing the Salesforce platform, ensuring accuracy and accessibility for streamlined operations. Additionally, automated data entry tasks using Java-based web scraping tools to efficiently extract, process, and input information from various online sources
- Utilized Microsoft Visio to design a variety of customizable cabinet templates tailored to meet the specific needs of customers

PROJECTS

Image Segmentation and Clustering

July – September 2021

- Implemented unsupervised clustering algorithms including PCA, Mean Shift, K-Means, and GMM for image segmentation
- Proficiently conducted data preprocessing, dimensionality reduction, and cluster analysis techniques to achieve accurate segmentation results
- Demonstrated expertise in computer vision and advanced clustering methods, contributing to the advancement of image analysis research

Handwriting Recognition

September – December 2019

- Developed a Python algorithm for handwritten character recognition, achieving 90% accuracy rate
- Utilized computer vision libraries, including OpenCV for image preprocessing and scikit-image for feature extraction. Implemented k-Nearest Neighbors (k-NN) for classification, combined with clustering techniques to enhance feature filtering and pattern recognition
- Leveraged pandas and numpy for data manipulation and analysis, optimizing the algorithm's performance and reliability

Hardware Sensors iOS App

September – December 2019

- Developed an iOS application using Swift that utilized CoreMotion to capture and process telemetry data from the phone's accelerometer, altimeter, magnetometer, and gyroscope
- Designed and implemented a dynamic UI/UX that displays live telemetry data, providing real-time insights and improving user interaction with data visualization

EDUCATION

Northeastern University | *M.S. in Electrical & Computer Engineering*

December 2023

Concentration in Hardware and Software for Machine Intelligence, Magna Cum Laude

University at Buffalo | *B.S. in Electrical Engineering*

May 2020

Minor in Mathematics, Cum Laude